

George Gu

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EDUCATION

University of Michigan

Bachelor of Science in Engineering, Computer Science

May 2028

Ann Arbor, MI

- **GPA:** 3.81/4.00
- **Coursework:** Operating Systems, Web Systems, Computer Security, Machine Learning, Artificial Intelligence, Reinforcement Learning, Theory of Computation, Computer Organization

EXPERIENCE

Capital One

Technology Intern

June – August 2026

Richmond, VA

- Built the **PUT /certify-asset** endpoint, one of 44 REST endpoints, behind a three-tier authorization model, shipping the endpoint alongside React badge components to production at **98% coverage** against an 80% gate.
- Architected **2 AWS EventBridge**-scheduled Python Lambdas: a daily certification-reminder digest querying PostgreSQL views to email owners, and a ServiceNow reconciliation pipeline uploading encrypted CMDB reports to S3.
- Authored idempotent **Flyway PostgreSQL** migrations building a ServiceNow CMDB caching layer (2 tables, 2 views, 2 indexes) within a 52-migration schema history, optimizing cross-table SQL for automated compliance reporting.

BoilerVault

Founding Engineer (Contract)

January 2026 – Present

West Lafayette, IN

- Built a multi-tenant operations platform with FastAPI, PostgreSQL, and Next.js/TypeScript for a **3-campus** student storage business, implementing JWT auth and 20+ REST endpoints validated by **140+ tests**.
- Reconciled out-of-order payment events across **4 Stripe accounts** and **3 WordPress booking sites** via idempotent webhook pipelines, automating customer, order, and billing creation with 0 manual entry across **\$130K+** in gross revenue.
- Migrated **2,000+** legacy records via three-tier matching (exact email, rapidfuzz fuzzy name, Stripe charge fallback), replacing a failing Zapier workflow and deploying on Railway/Vercel as the production system of record.

Nexteer Automotive

Software Engineering Intern

May – August 2025

Saginaw, MI

- Developed an **LLM-powered** IDE extension in Python and TypeScript that parsed **300+** internal engineering guidelines and automated compliance checks across C and H files, integrating Azure AI and Copilot APIs.
- Designed prompt pipelines and few-shot strategies to extract rules from unstructured documentation, tuning the system to **95% violation-detection accuracy** and saving engineers an estimated 8 hours of review per week.
- Deployed the tool across **26 sites** and integrated it with internal dev pipelines, cutting manual code-review time by **87.5%** for safety-critical embedded steering systems.

Villanova University

Data Engineer & First Author

June – September 2023

Villanova, PA

- Engineered end-to-end data pipelines in R, using PCA and clustering to process and visualize **28,000+** pathogen isolates across **10+ years**, integrating bacterial species, antimicrobial agents, and AMR resistance genes to uncover essential trends.

PROJECTS

Scalable Search Engine

Systems Developer

January – April 2026

Ann Arbor, MI

- Built a multi-stage MapReduce pipeline in Python to construct an inverted index over **3,000+ documents**, computing TF-IDF scores and normalization factors across a multi-job processing architecture for scalable retrieval.
- Implemented a REST API index server in Flask serving **3 partitioned inverted index segments**, integrating PageRank-weighted cosine similarity to balance TF-IDF with link authority to rank query results by relevance.
- Designed a service-oriented search frontend that dispatches concurrent REST requests across segment servers using threads, then aggregates and re-ranks the results to return the **top 10** hits.

Resume Screener

Machine Learning Developer

January – May 2025

Ann Arbor, MI

- Built an **NLP** pipeline over a labeled dataset of **1,000+ resumes and job postings**, applying tokenization, TF-IDF vectorization, and feature engineering for resume-to-posting matching.
- Designed **scikit-learn** and **TensorFlow** models to rank resumes against job postings, using sentence-transformer embeddings and **RAG** retrieval over job descriptions, deployed via Streamlit to **100+ users**.

SKILLS/ADDITIONAL

Languages: Python, SQL, C/C++, Java, JavaScript, TypeScript, HTML, CSS

Frameworks/Developer Tools: Git, Docker, Kubernetes, Unix, Node.js, React.js, Flask, PyTorch, TensorFlow, PostgreSQL

Interests: Philosophy and Metaphysics, Game Design, Boulderling, Skateboarding, Golf, Guitar, Calisthenics

PUBLICATION

Gu G, Pei H, Zhou A, Fan B, Zhou H, Choi A, Huang Z, “A Comprehensive Study of Historical Detection Data for Pathogen Isolates from U. S. Cattle”, 2023, *Antibiotics*, 12, 1509.

doi: <https://doi.org/10.3390/antibiotics12101509>